



Predicting Entrepreneurial Activity Using Machine Learning Techniques

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Highlights

- Analysis of entrepreneurial activity prediction using supervised machine learning models
- Comparison of models: XGBoost, CatBoost, Random Forest, Decision Trees, H2O GBM, ANN, k-NN, Naive Bayes, Lasso, Ridge, ElasticNet and Logistic Regression
- Comparative Analysis of pooled and yearwise NME and OME data for Indian and World Context.

Abstract

This Study analyses the predictability of entrepreneurial activities using a number of supervised machine learning models that include XGBoost, CatBoost, Random Forest, Decision Trees, H2O GBM, ANN, k-NN, Naive Bayes, Lasso, Ridge, ElasticNet and Logistic Regression on the Global Entrepreneurship Monitor dataset.

These ML models were compared to evaluate their effectiveness in predicting opportunity-motivated entrepreneurship(OME) and necessity-motivated entrepreneurship(NME).

Across pooled data, XGBoost was a consistently strong performer in AUC for World Context, while Random Forest was a strong performer for Indian Context. To ensure no outliers were skewing results, a detailed year-wise analysis was conducted.

Keywords:

Entrepreneurship, Machine Learning, SMOTE, H2O GBM, XGBoost, Random Forest, Global Entrepreneurship Monitor.

1. Introduction

Entrepreneurial activity is generally viewed as a major force behind economic growth and innovation. In the continuously evolving world, demanding new innovations, new services, and the demand for ease of lifestyle has surged the demand for entrepreneurship. Nonetheless, predicting the phenomenon of entrepreneurship is still a tough challenge owing to its intricate and multi-faceted character. Previous studies have primarily depended on the use of linear statistical models, which take for granted fixed and simplified interactions among the variables. Often, non-linear and reciprocal factors that affect entrepreneurial decisions are simply not characterised (Bruyat & Julien, 2001; Churchill & Bygrave, 1990). Therefore, there is slowly but surely a growing interest in applying analytical techniques that can manage high-dimensional data and flexible relationships.

Machine learning techniques (ML) allow models to learn to recognise complex patterns without the use of predetermined functional forms. This ability gets even more important when it comes to deciphering opportunity-motivated entrepreneurial activity (OME), because it is governed by individual traits, socio-economic conditions and the general context. Recent research indicates that ML methods are becoming increasingly relevant in the area of predictive accuracy enhancement in entrepreneurship studies (Schade & Schuhmacher, 2023).

This research includes the use of modern ML techniques: Lasso, Ridge, ElasticNet, CatBoost, H2O GBM(Gradient Boosting Model), besides benchmark models like XGBoost and ANN, and employs them on the Global Entrepreneurship Monitor (GEM) dataset, which consists of 1,016,719 individuals from 85 countries.

The goals of this research study are as follows:

- 1) Evaluate the accuracy of predictions produced by advanced ML methods in comparison with classical models
- 2) Treat the problems resulting from data imbalance by utilising conventional oversampling methods like SMOTE.
- 3) Reveal the significant factors that influence the occurrence of entrepreneurs motivated by opportunities.

By accomplishing these goals, this research not only plays a part in the methodological advancement of studies related to entrepreneurship but also provides trustworthy information that can lead to the formulation of policies regarding opportunity-driven entrepreneurship.

2. Literature Review

Several studies have examined the prediction of entrepreneurial activity across different countries. Most of the early literature concentrated on linear relationships between macroeconomic indicators and the growth of entrepreneurial activity. However, these studies—largely based on simple linear regression—consistently struggled to establish strong or stable correlations between macroeconomic factors and entrepreneurial behaviour ([Castaño et al., 2015](#)).

Recent research has shifted toward non-linear modelling and machine learning (ML) approaches. For instance, [Schade and Schuhmacher \(2023\)](#) applied machine learning techniques such as regression trees, artificial neural networks, k-nearest neighbours, and naïve Bayes to predict entrepreneurial activity. Their study reported accuracy rates ranging from 70.1% to 91.2% for Opportunity-Motivated Entrepreneurship (OME), identifying extreme gradient boosting (XGBoost) as the best-performing model. For Necessity-Motivated Entrepreneurship (NME), they achieved an accuracy of 96.5%, with XGBoost again outperforming all other algorithms.

Similarly, [Koumbarakis et al. \(2023\)](#) used multiple ML models to forecast firm birth and abandonment over the first five years of business development. Their work compared traditional logistic regression with random forests, artificial neural networks, k-nearest neighbours, and gradient boosting. While extreme gradient boosting demonstrated strong overall performance, neural networks were particularly effective in predicting early-stage entrepreneurial abandonment.

[Zhang et al. \(2022\)](#) focused on predicting entrepreneurial intentions among students by integrating an improved crow search algorithm with a kernel extreme learning machine and feature selection (CSA-KELM-FS). Using data from 842 students, their model achieved an accuracy of 93.2% and displayed strong predictive stability.

In another study, [Virasa et al. \(2022\)](#) examined entrepreneurial intentions across six ASEAN countries—Vietnam, the Philippines, Indonesia, Malaysia, Thailand, and Singapore—using the Theory of Planned Behaviour (TPB). Drawing on 13,358 responses from the Global Entrepreneurship Monitor (GEM), the authors highlighted significant behavioural and contextual drivers of entrepreneurial intention.

Building on this strand of research, [Castaño et al. \(2015\)](#) analysed how social, cultural, and economic conditions influence entrepreneurial outcomes, further emphasising the role of complex non-linear dynamics in entrepreneurial ecosystems.

Despite these contributions, a clear gap persists in the comparative evaluation of machine learning models for entrepreneurship prediction. Many studies rely on accuracy as the primary performance metric, while overlooking complementary indicators such as the Area Under the Curve (AUC), which offers a more robust measure of class separability. Moreover, although XGBoost frequently emerges as a leading model, the literature has not sufficiently explored side-by-side comparisons with emerging AutoML frameworks such as CatBoost or H2O's Gradient Boosting Machine that may offer competitive or superior performance.

As machine learning techniques continue to evolve, there is substantial scope for further research into the predictive power, interpretability, and generalisation ability of these advanced models.

3. Data

The study uses the Adult Population Survey individual-level dataset sourced from the Global Entrepreneurship Monitor(GEM) database.

The APS dataset is a cross-sectional sample of 1,016,719 individuals' survey responses from 85 countries for 5 consecutive years, 2015-2020. The survey dataset captures the characteristics, motivations and ambitions of individuals starting businesses, as well as social attitudes towards entrepreneurship.

The key variable in this study is binary classification of opportunity-motivated early-stage entrepreneurship, which is represented by the TEAyyOPP measure for the years 2015–2017 and the updated motivator variables teayymot1yes and teayymot2yes for the years 2019–2020. Additionally, various explanatory variables were selected based on the established literature on entrepreneurial behavior.

These include Human capital Features like Educational status, employment status, and income levels. Psychological Traits: Entrepreneurial self-efficacy (the belief in one's ability to start a business), fear of failure. Socio-Demographic Characteristics: Gender, age, household size, and whether an individual knows an entrepreneur.

Contextual Factors: Country-specific variables, such as the developmental stage of the country

The dataset is comprehensive and consists of different demographic characteristics, human-capital indicators, psychological factors such as self-efficacy and fear of failure, and employment-related aspects.

Prior to modelling, preprocessing steps were added to including the removal of missing data, encoding of categorical variables like gender, household size into numerical binary form, etc.

Ensuring consistent variable definitions for different years and creating a reliable dataset for machine-learning classification were the main objectives of the data preparation process.

According to the previous GEM research, the variables occurt and occuds were merged into one variable, occurrd, which indicates that the respondents were retired or disabled and thus stopped working.

After cleaning the data across survey years, a total of 21 variables were included in the analysis.

Table 1: NME Global Descriptive Statistics(2015-2020)

No.	Variable (Meaning)	Obs	Mean	Std. Dev.	Min	Max
1	Country Code	1,016,719	220.013	290.601	1	995
2	Respondent ID (SETID)	1,016,719	3.44×10^{12}	1.41×10^{13}	1.15×10^9	9.74×10^{13}
3	Survey Year	1,016,719	2017.349	1.684	2015	2020
4	Gender (1=Male, 2=Female)	1,016,710	1.498	0.500	1	2
5	Age	996,757	41.380	14.607	18	99
6	Household Size	998,327	97.369	93,634.28	0	93,600,000
7	Household Income Category (GEMHHINC)	826,674	23,531.01	31,238.47	33	68,100
8	Educational Attainment (GEMEDUC)	1,001,001	1000.429	544.765	0	1,720
9	Perceived Opportunities (oport)	882,764	1.235	1.487	0	5
10	Perceived Skills (suskill)	982,264	1.326	1.569	0	5
11	Fear of Failure	978,161	1.228	1.544	0	5
12	Media Attention to Entrepreneurship (nbmedia)	891,197	1.425	1.520	0	5
13	Future Support for Entrepreneurship (futsup)	965,527	0.228	0.420	0	1
14	Full-Time Work (occufull)	1,010,047	1.572	0.495	1	2
15	Part-Time Work (occupart)	1,008,910	1.881	0.324	1	2
16	Job Seeking (occuseek)	606,361	1.810	0.393	1	2
17	Student (occustu)	609,196	1.859	0.348	1	2
18	Home Duties (occuhome)	512,564	1.698	0.459	1	2
19	Knowing an Entrepreneur (knowent)	996,209	0.424	0.494	0	1
20	TEA – Necessity Entrepreneurship (TEAyyNEC)	1,016,719	0.047	0.211	0	1

21	Business Ownership (occurd)	512,189	0.531	0.756	0	2
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Table 2: OME Global Descriptive Statistics(2015-2020)

No.	Variable (Meaning)	Obs	Mean	Std. Dev.	Min	Max
1	Respondent ID (SETID)	1,016,719	3.44×10^{12}	1.41×10^{13}	1.15×10^9	9.74×10^{13}
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12	Full-Time Employment (occufull)	1,010,047	1.572	0.495	1	2
13	Part-Time Employment (occupart)	1,008,910	1.881	0.324	1	2
14	Job-Seeking Status (occuseek)	606,361	1.810	0.393	1	2
15	Student Status (occustu)	609,196	1.859	0.348	1	2
16	Home Duties (occuhome)	512,564	1.698	0.459	1	2
17	Business Ownership (occurd)	442,099	0.616	0.782	0	2
18	Gender (1=Male, 2=Female)	1,016,710	1.498	0.500	1	2
19	Age	996,757	41.380	14.607	18	99
20	Household Size (hhsiz)	998,327	97.369	93,634.28	0	93,600,000

21	Opportunity-Driven TEA (TEAyyOPP)	1,016,719	0.083	0.276	0	1
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4. Methods

4.1 Data Sampling

The GEM dataset was divided into two parts: a "training and validation" sample and a separate "hold-out" test sample, as per the method suggested by Mullainathan and Spiess 2017. For the machine learning models training and validation, 70% of the data was used. The rest of the 30% was set aside for testing the models' predictive performance on the data that had not been seen before, in accordance with the suggestion of (Choudhury et al. 2021).

For a balanced entrepreneurship dataset, which is required for ML models, the probability of engaging and not engaging in OME and NME should be equal.

However, due to entrepreneurial activity being uncommon, the survey dataset is imbalanced with fewer instances of engaging in OME (i.e., 1 = yes) compared to not engaging (i.e., 0 = no). This issue is mitigated by resampling the dataset using SMOTE (synthetic minority over-sampling technique). SMOTE generates synthetic minority-class samples by interpolating between existing minority observations, helping reduce class imbalance and improve classifier performance. ([Chawla et al. 2002](#)).

In order to verify that the models had indeed a good predictive performance for new data, we used a 10-fold cross-validation method. To clarify, in this method, the training dataset is divided randomly into k parts that are almost equal.

This means that every part will be used as a validation set only once (which is similar to a hold-out sample), while the remaining $k - 1$ parts will be used to train the model. This way, a fair measurement of predictive accuracy is possible ([Schade & Schuhmacher, 2023](#)). In the present research, we decided on a 10-fold cross-validation scheme ($k = 10$), which is frequently suggested in machine-learning literature ([Choudhury et al., 2020](#)).

4.2 Machine Learning Models

To predict OME (opportunity and NME, this study utilizes multiple supervised machine learning techniques on default hyperparameter settings as indicated in Table 2 to ensure direct comparison without human intervention([Schade & Schuhmacher, 2023](#)):

1. Logistic Regression: Logistic Regression gives the likelihood of a binary outcome by a logistic function applied to the linear predictors. It is used as a baseline to compare with the other ML models([Schade & Schuhmacher, 2023](#)).
2. Artificial Neural Network (ANN): ANNs cut the corner of complex non-linear connections through grouped layers of computing units.
3. k-Nearest Neighbours (kNN): kNN assigns a class to the instance based on the most frequent class among its nearest neighbours in the feature space (Cover & Hart, 1967).
4. XGBoost: XGBoost is a gradient boosting model that is highly optimised for both speed and accuracy using regularised boosting methods (Chen & Guestrin, 2016).
5. Decision Tree: Decision Trees sort the data by first splitting the features according to conditions that give the most information gain (Quinlan, 1986).
6. Naive Bayes: Naive Bayes is a classifier based on probability that uses Bayes' theorem under independence assumptions (McCallum & Nigam, 1998).
7. Lasso: The Lasso model imposes an L1 penalty, leading to variable selection and regularisation, which in turn makes the model more accurate in predicting outcomes in high-dimensional situations (Tibshirani, 1996).
8. Ridge Regression: Ridge applies an L2 penalty to lower the variance of the coefficients and to solve multicollinearity in regression Issue (Hoerl & Kennard, 1970).
9. Elastic Net: Elastic Net is a combination of L1 and L2 penalties, which makes the method robust when there are correlations among the predictors (Zou & Hastie, 2005). Random Forest: Random Forest is a group of decision trees which gives a new prediction by taking the average of the predictions given by multiple bootstrapped trees (Breiman, 2001).
10. CatBoost: CatBoost is an uprising gradient boosting algorithm for the efficient handling of categorical features and less prediction bias (Prokhorenkova et al., 2018).

Table 2: Default Hyperparameters

Model	Hyperparameters
Lasso	alpha = 1, lambda = 0.01
Ridge	alpha = 0, lambda = 0.01
Elastic Net	alpha = 0.5, lambda = 0.01
Random Forest	ntree = 100
ANN	size = 5, decay = 0.1, maxit = 100
kNN	k = 5, weights = “prob”
XGBoost	nrounds = 100, learning_rate = 0.3
Decision Tree	cp = 0.01, minsplit = 20, minbucket $\approx$ 7, maxdepth = 30, xval = 10
Logistic Regression	binomial logit, threshold = 0.5
Naive Bayes	laplace = 0, usekernel = TRUE, adjust = 1, kernel density for numeric features
CatBoost	iterations = 100, learning_rate = 0.1
H2O GBM	ntrees = 100, learn_rate = 0.1

5. Results

5.1 Pooled Data(2015-2020)Performance

In this section, we demonstrate the predictive performance of the supervised machine learning models that were applied to the OME and NME datasets not only in the Indian context but also in the Global context using SMOTE as the resampling method. The GEM dataset serves as the source for all results.

OME – Global Context (2015–2020)

XGBoost was the best performer on the global OME dataset. It produced:Accuracy: 90.11% AUC: 0.8539

Random Forest and CatBoost achieved similar levels of super accuracy (89–90%) but not enough recall due to complex opportunity-driven entrepreneurial distribution.

Conventional models (Lasso, Ridge, Elastic Net) always gave accuracies of 73–74% and very poor precision (~0.25) which indicated their inability to capture complex nonlinearities. On the other hand, kNN was the lowest in the performance on the overall metrics.

Thus, the results of OME prediction at the global level indicates the use of the gradient-boosting methods, with XGBoost being the best among them.

TABLE 3: OME WORLD 2015-2020 PERFORMANCE METRICS

Model	Accuracy	Recall	Precision	F1_Score	AUC
Lasso	0.7366024	0.7816564	0.2524315	0.3816207	0.8261083
Ridge	0.7405167	0.7789185	0.2550998	0.3843296	0.8280639
Elastic Net	0.7389510	0.7816564	0.2542864	0.3837366	0.8275139
Random Forest	0.8974450	0.1704312	0.5209205	0.2568334	0.8345691
ANN	0.1039784	1.0000000	0.1039784	0.1883703	0.5000000
kNN	0.7222262	0.4818617	0.1828571	0.2651101	0.6521543
XGBoost	0.9011458	0.2176591	0.5638298	0.3140741	0.8538682
Decision Tree	0.8101203	0.5222450	0.2791804	0.3638531	0.7497834
Logistic Regression	0.7422248	0.7754962	0.2559295	0.3848505	0.8283923
Naive Bayes	0.7240766	0.7796030	0.2426502	0.3701056	0.8082466
CatBoost	0.8911821	0.2320329	0.4544236	0.3072043	0.8376143
H2O GBM	0.8959505	0.2621492	0.4993481	0.3438061	0.8524625

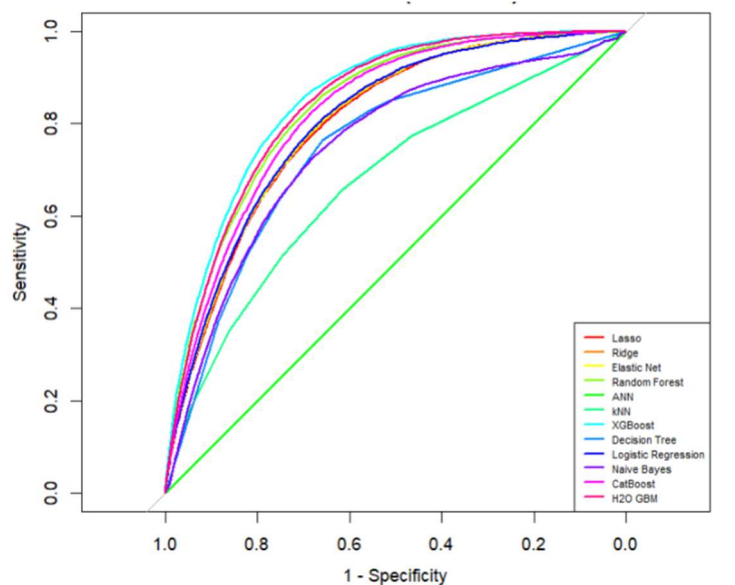


Fig 1. ROC curves comparing performance of all ML models on OME Global Dataset(2015-2020)

OME – Indian Context (2015–2020)

The performance of the classifiers in the Indian context was similar to the global findings. The Random Forest and XGBoost models achieved the highest accuracies, respectively: Random Forest: (Accuracy = 0.9046, AUC = 0.8367) and XGBoost:(Accuracy = 0.9025, AUC = 0.8201)

Regularized linear models exhibited a performance range of mid-point (accuracy ~0.76-0.77, AUC ~0.83). The results suggest that the OME prediction for India also favors the tree model ensembles to the greatest extent, with Random Forest giving the best compromise between sensitivity and overall predictive power.

TABLE 4: OME INDIA 2015-2020 PERFORMANCE METRICS

Model	Accuracy	Recall	F1_Score	AUC
Lasso	0.7646070	0.7510730	0.3846154	0.8294961
Ridge	0.7692308	0.7296137	0.3824522	0.8303961
Elastic Net	0.7696511	0.7510730	0.3897550	0.8304461

Random Forest	0.9045818	0.2317597	0.3223881	0.8366869
ANN	0.9020597	0.0000000	NA	0.5000000
kNN	0.6515343	0.3733906	0.1734796	0.5440994
XGBoost	0.9024800	0.2274678	0.3136095	0.8200765
Decision Tree	0.8440521	0.4721030	0.3722504	0.7324176
Logistic Regression	0.7713325	0.7381974	0.3873874	0.8309461
Naive Bayes	0.7633459	0.6437768	0.3476246	0.7844558
CatBoost	0.8995376	0.2789700	0.3523035	0.8360019
H2O GBM	0.8949138	0.3090129	0.3654822	0.8395998

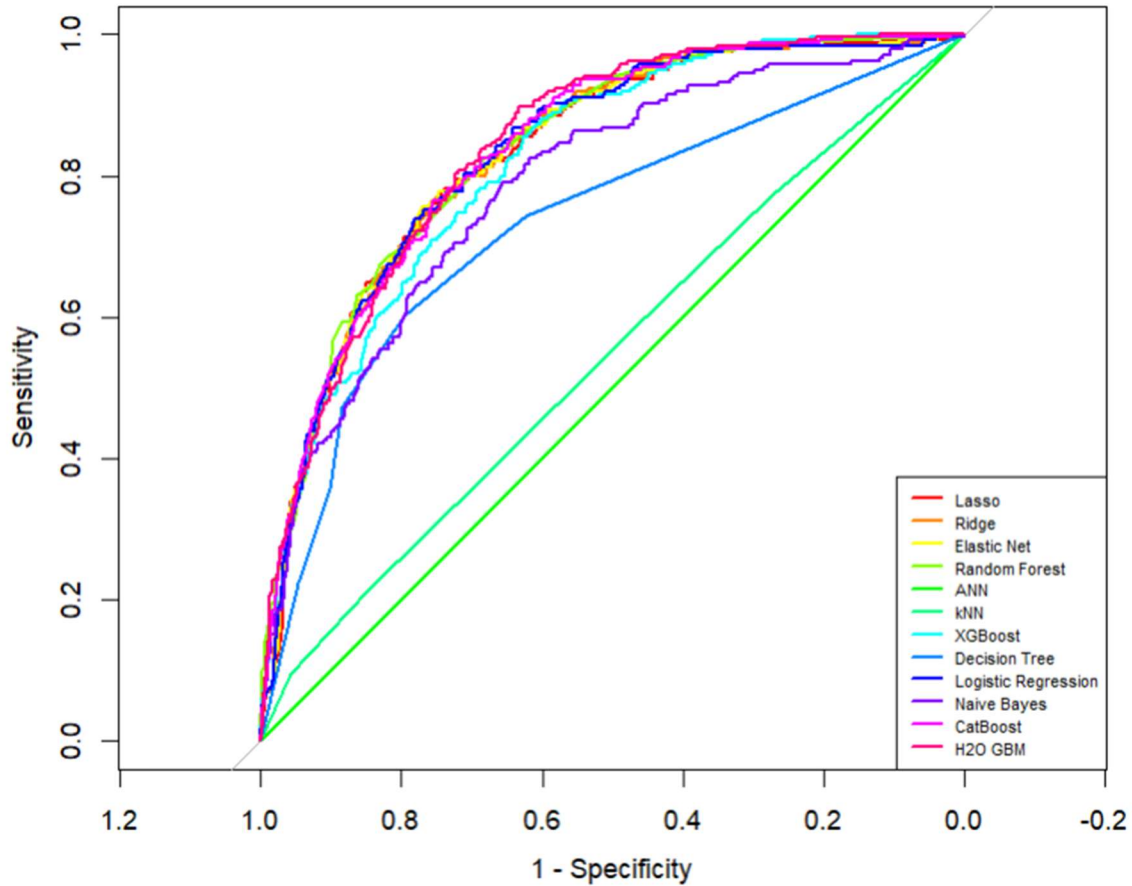


Fig 2. ROC curves comparing performance of all ML models on OME Indian Dataset(2015-2020)

NME Indian Context (2015–2020)

In the NME dataset for India, the application of ensemble learning techniques to linear models was clearly superior. The Random Forest model not only reached the maximum overall accuracy of 94.30% but also garnered the highest recall of 98.69%, confirming its robustness in identifying necessity-motivated entrepreneurs. The CatBoost and XGBoost also showed up closely with their accuracies of 94.01% and 93.56%, respectively. On the other hand, Lasso, Ridge, and Elastic Net which are models based on linear algorithms, achieved around 83-84% accuracies with somewhat similar recall values (~84-85%) as well. The Decision Tree classifier delivered a good performance (88.76% accuracy), while kNN and Naive Bayes demonstrated weakness in comparison. The ANN model exhibited perfect recall (100%) but, at the same time, extremely poor discrimination ($AUC = 0.50$), which indicates that it was overfitting under SMOTE sampling.

Hence, the findings indicate that tree-based ensembles are still the most dependable method for predicting NME in the Indian scenario.

TABLE 5: NME WORLD 2015-2020 PERFORMANCE METRICS

Model	Accuracy	Recall	F1_Score	AUC
Lasso	0.8340870	0.8450966	0.3402490	0.8343787
Ridge	0.8344348	0.8465825	0.3351955	0.8329740
Elastic Net	0.8354783	0.8469539	0.3403068	0.8344092
Random Forest	0.9429565	0.9869985	0.3970588	0.8493847
ANN	0.9363478	1.0000000	NaN	0.5000000
kNN	0.7036522	0.7232541	0.1513944	0.4480813
XGBoost	0.9356522	0.9795691	0.3642612	0.8447474
Decision Tree	0.8876522	0.9182764	0.3312629	0.7949764
Logistic Regression	0.8351304	0.8465825	0.3398329	0.8351724
Naive Bayes	0.7829565	0.7945765	0.2641509	0.7940264
CatBoost	0.9401739	0.9832838	0.3943662	0.8480582
H2O GBM	0.9304348	0.9650817	0.4350282	0.8624766

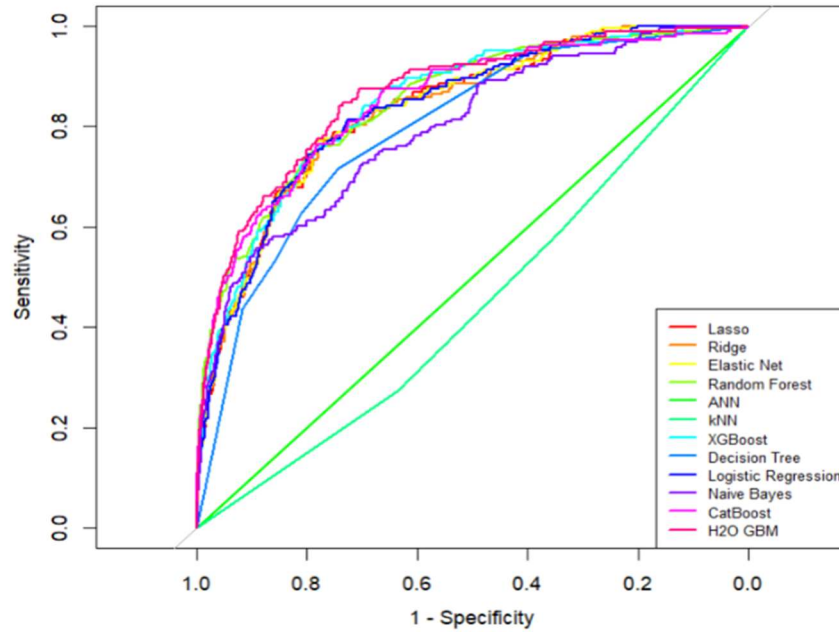


Fig 3. ROC curves comparing performance of all ML models on NME Indian Dataset(2015-2020)

NME – Global Context (2015–2020)

The performance trends worldwide are different. High recall (near 100%) is recorded for the Random Forest, Decision Tree, ANN, and CatBoost models, but their F1-scores were low because the class imbalance severely affected the results and could not be mitigated by the SMOTE sampling.

The regularized linear models (Lasso, Ridge, Elastic Net) gave stable yet moderate predictive quality, garnering 86%–87% accuracy levels with 0.17 F1-scores. XGBoost was better than usual ensemble models with regard to AUC (0.778), as it exhibited more pronounced discrimination capacity. XGBoost is considered globally the most balanced NME predictor, as the AUC values confirming this were obtained.

TABLE 6: NME WORLD 2015-2020 PERFORMANCE METRICS

Model	Accuracy	Recall	F1_Score	AUC
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Lasso	0.8701059	0.8946530	0.17355794	0.7591645
Ridge	0.8648107	0.8883674	0.17523250	0.7605121
Elastic Net	0.8679397	0.8918874	0.17617618	0.7597556
Random Forest	0.9573171	0.9996648	0.01481481	0.7366022
ANN	0.9573171	1.0000000	NaN	0.5000000
kNN	0.8583119	0.8853503	0.13176008	0.5876157
XGBoost	0.9567555	0.9984915	0.03921569	0.7784855
Decision Tree	0.9573171	1.0000000	NaN	0.7100897
Logistic Regression	0.8565469	0.8786457	0.17679558	0.7601536
Naive Bayes	0.6726573	0.6723936	0.15035402	0.7351444
CatBoost	0.9573171	1.0000000	NaN	0.7736209
H2O GBM	0.9523427	0.9919544	0.10271903	0.7935296

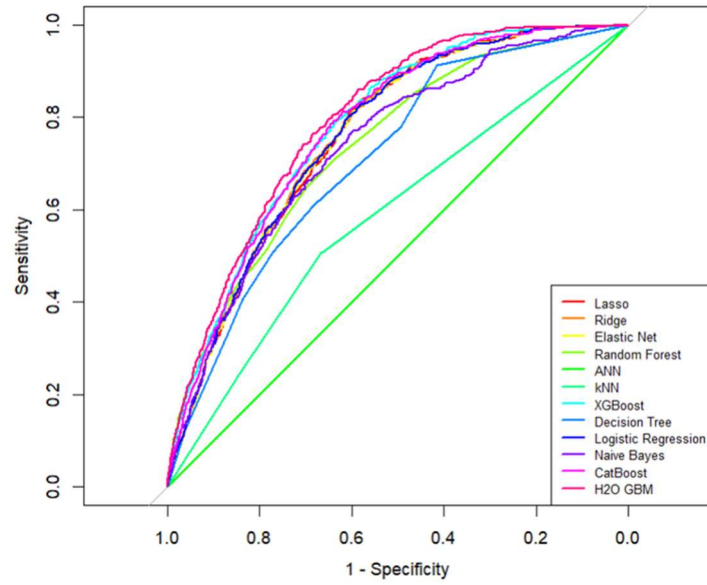


Fig 4. ROC curves comparing performance of all ML models on NME Indian Dataset(2015-2020)

5.2 Comparison of Year-Wise Performance with Pooled Results

The pooled 2015-2020 results provide an overall picture of which machine learning models perform best when all years are combined. But, combining 6 years of GEM data can mask yearly fluctuations in entrepreneurial behaviour.

Therefore, the year-wise results(2015-2020) are compared with pooled results to assess model stability and dataset volatility. To illustrate the yearwise comparison, the model with the best AUC was selected and its corresponding accuracy has been represented alongside in graphical format to measure accuracy of the results. Year-wise tables are provided in Appendix.

OME – Global Context

XGBoost was determined as the most powerful model in the pooled global OME results with AUC=0.8539 and accuracy=90.11%; H2O GBM and CatBoost were the next in line. The superior performance of models based on non-linear approaches with powerful gradient-boosting capability indicates that global opportunity-driven entrepreneurship favors such models.

Year-wise results strengthen the conclusion but also indicate replacement between XGBoost and H2O GBM.

H2O GBM performs best in the years 2016, 2018, and 2020 while XGBoost is best in 2015, 2017, and 2019.

This replacement does not reflect volatility but rather the situation of two very similar boosting technologies responding to minute shifts in global economics.

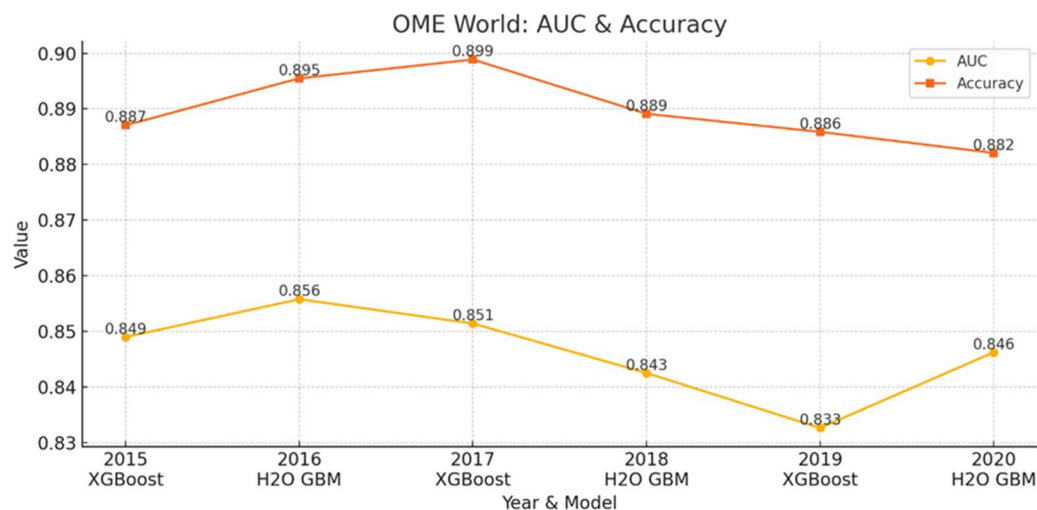


Fig 5. Yearwise analysis of model with best AUC and corresponding accuracy for OME Global Dataset(2015-2020)

OME- Indian Context

The pooled OME results for India validate Random Forest as the best model with the highest accuracy (0.9046, AUC = 0.8367).

The annual comparison, however, points to a lot of ups and downs, exposing the fact that opportunity-motivated entrepreneurship in India does not have a steady course. Over the years, Random Forest has been the best model in the high-opportunity years of 2015, 2017, and 2018, with 2017 exhibiting the highest year-wise AUC of 0.9660.

On the other hand, the linear models have been superior to the tree-based ones in several occasions during the same period; Ridge was the best in 2016 and 2019, while Lasso was the best in 2020. Such results imply that the drivers of opportunity-based Entrepreneurship change over time, sometimes giving rise to complex non-linear patterns and at other times being expressed as more linear relationships. Hence, although pooled results proclaim Random Forest as the long-run leader, the year-wise results portray a dynamic and cyclical entrepreneurial landscape that pooled aggregation has obscured.

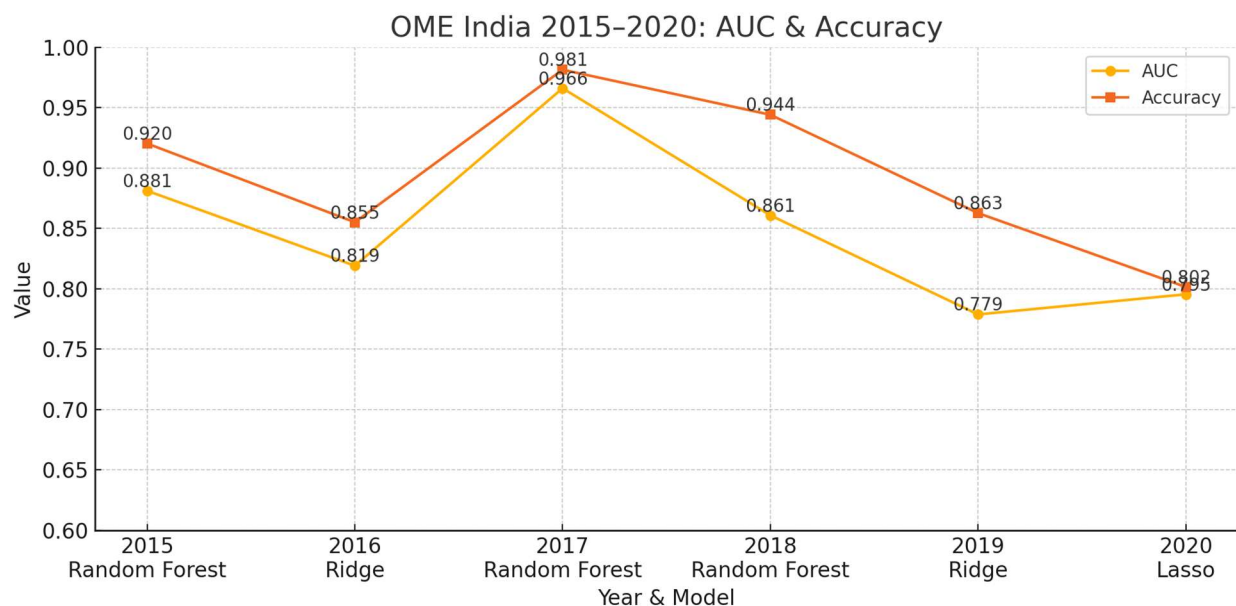


Fig 6. Yearwise analysis of model with best AUC and corresponding accuracy for OME Indian Dataset(2015-2020)

NME- World Context

The combined global NME results show that there are several ensemble methods that reach very high recall (almost 100%), but H2O GBM gives the strongest overall balance, especially with its AUC of 0.7935, followed closely by XGBoost.

If we look at the performance year-wise, this trend is still mainly true, with H2O GBM being the top model in most years: 2016, 2017, 2018, and 2020 ; whereas XGBoost takes the lead in 2015 and 2019. Global annual performance shows only slight variations unlike Indian results, and the winning models still stay among the same pair of boosting algorithms.

Such a consistent pattern might point out that necessity entrepreneurship on a global scale is characterized by macro-patterns that take time to change rather than sudden shifts.

Thus, the pooled performance is truly reflective of global NME activity and does not mask any substantial yearly differences. The correspondence between pooled and yearly patterns is an indication of the structural stability of necessity entrepreneurship amongst countries.

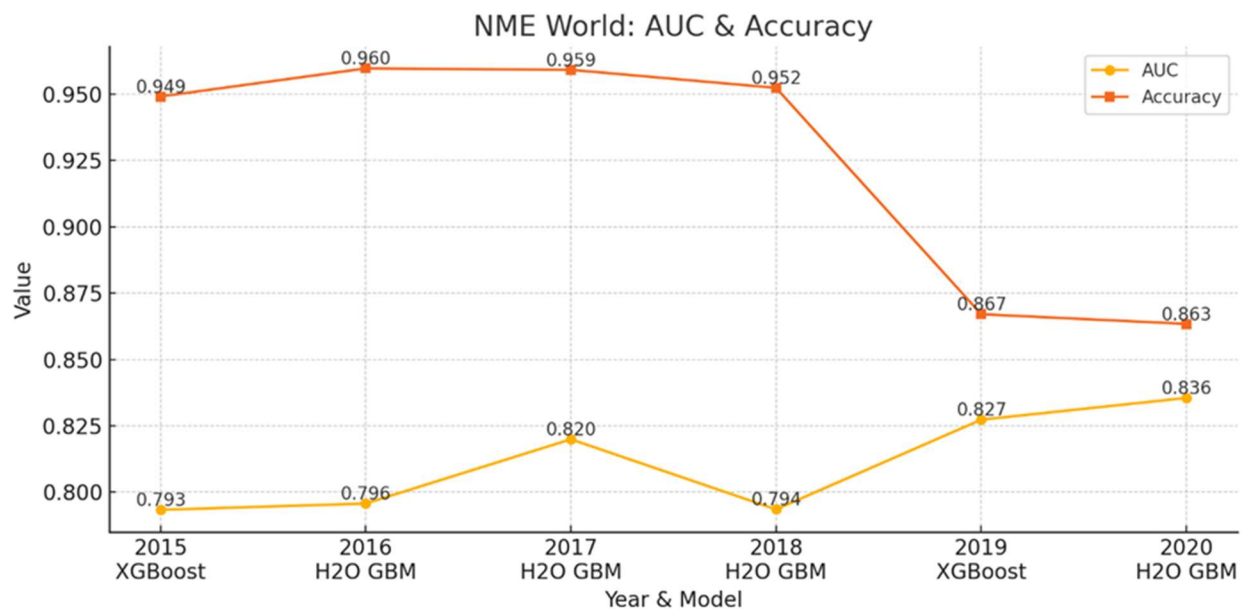


Fig 7. Yearwise analysis of model with best AUC and corresponding accuracy for NME World Dataset(2015-2020)

NME Indian Context

In the pooled NME results for India, ensemble models, especially Random Forest, have shown a steady and substantial increase in performance, obtaining the highest figures for accuracy (94.30%) and recall (98.69%), with CatBoost and XGBoost closely trailing behind. These findings imply that if data from all six years is combined, tree-based methods will consistently be the most powerful ones.

On the contrary, the year-wise analysis uncovers a great deal of variability that the pooled results do not reveal. Among the three models, CatBoost is the top-ranked in 2015(82.66%), 2016(85.75%), and 2020(80.67%); while the other two-year periods especially, Naive Bayes is the top model with the highest AUC in both 2017(86.30%) and 2018(73.8%), and Ridge Regression(77.36%) is the best in 2019.

This variation signifies that the framework of necessity-motivated entrepreneurship is radically different every year. Some years show complicated, non-linear behavior that is suitable for boosting models, while others conform to simple probabilistic or linear decision boundaries.

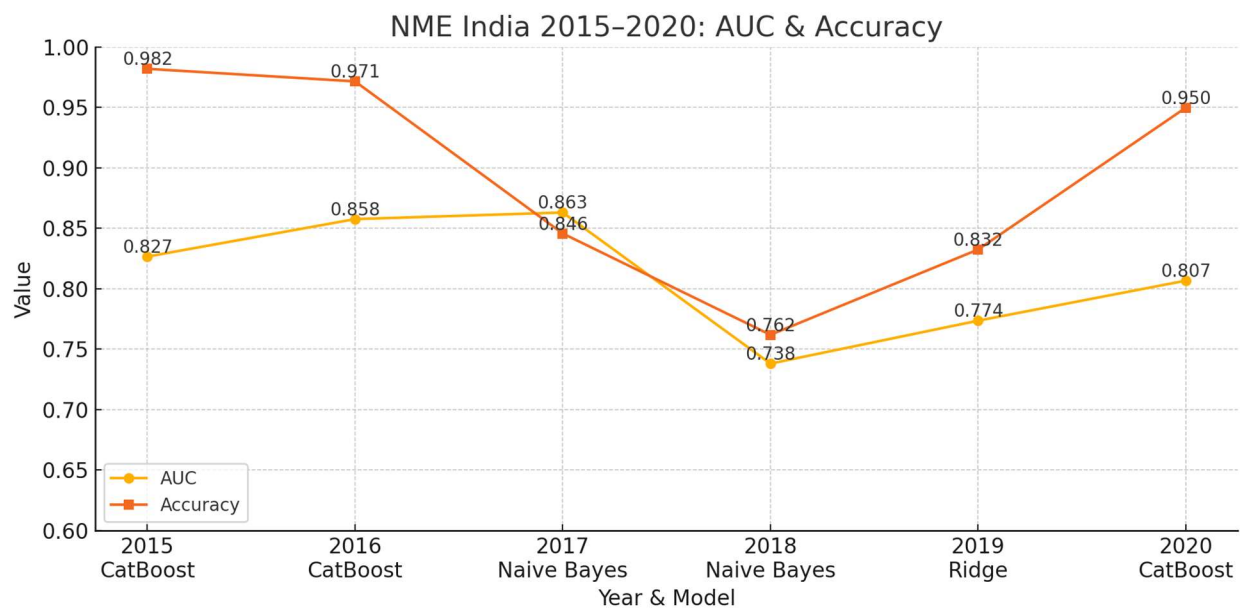


Fig 8. Yearwise analysis of model with best AUC and corresponding accuracy for NME Indian Dataset(2015-2020)

6. Discussion

6.1 Overview of Model Performance of Pooled Data

The comprehensive verdicts of this study reveal that, in general, tree-based ensemble modeling: Random Forest, XGBoost, CatBoost, and H2O GBM in particular, always yield better predictions in the OME and NME datasets not only in India but also globally.

In the combined results for 2015-2020, XGBoost is shown to be the best predictor of entrepreneurship worldwide, as it attained the greatest AUC and accuracy in Global OME and NME datasets, whereas Random Forest is ranked first in the Indian OME and NME datasets, thereby emphasizing its proficiency in modeling non-linear patterns that vary from one context to another.

Among the models such as Lasso, Ridge, and Elastic Net, the latter two provide relatively consistent but limited predictions. Conversely, the kNN classifier suffers from poor performance on both contexts, while the ANN model's performance is inconsistent with SMOTE applied, as the model either shows high recall or very low discrimination, indicating its susceptibility to data imbalance.

6.2 Overview of Year-Wise Analysis

The findings for each year provide valuable information about the changes in entrepreneurship over time that cannot be seen in the pooled results. In the case of NME India, the CatBoost model is the most accurate during the stable period of years, but Naive Bayes surprisingly takes the lead in the years 2017-2018 while the Ridge model performs best in the year 2019. This indicates a very strong year-to-year fluctuation in NME India. The accuracies also change a lot, particularly in the years where probabilistic or linear models are the most dominant ones, which may imply a shift in labor-market pressure, economic uncertainty, or domestic constraints. On the contrary to NME global which shows high stability: H2O GBM is the best model in most years and XGBoost only takes the lead in 2015 and 2019 with the accuracies remaining between 86 and 95 percent. This correlation indicates that the combined global outcomes are trustworthily depicting the yearly patterns.

In the case of OME India year-wise data, Random Forest is the top model in high-opportunity years (2015, 2017, 2018) with very high accuracies, while Ridge and Lasso are the front runners in slow or disruption-heavy years (2016, 2019, 2020) often with lower accuracy.

In contrast, OME Global is the most stable, with XGBoost and H2O GBM alternating in dominance but maintaining high accuracies and AUCs, thus, the pooled findings are roughly equal to yearly model performance in summarization.

All in all, the yearly comparison reveals performance of machine learning models on Indian dataset as being much more volatile on a year wise basis while global ones stay structurally consistent leading to the assumption that pooled models are more representative in the case of the global context.

6.3 Theoretical Inferences, Limitations and Future Improvements

The results hold substantial significance for theory as well as for practice. From a theoretical standpoint, the application of ensemble methods to the datasets yielded robust outcomes, which, in turn, gave support to the hypothesis that entrepreneurial motivation originates from multi-dimensional, interaction-heavy mechanisms rather than linear cause-effects. This is in line with the theories of behavioural and institutional entrepreneurship which consider that complexity, heterogeneity, and context dependency are the main characteristics of such phenomena. The varying performance in India year after year

corroborates the cyclical economic pressures, informal networks, and policies as interrelated factors in the entrepreneurship process.

The study also has several limitations that can provide direction for future research. On one hand, the necessity of SMOTE oversampling for the problem of class imbalance has the effect of recalling and possibly introducing synthetic noise, which will definitely play a role in affecting the performance of ANN and Naive Bayes. Therefore, the future studies might better consider the advanced imbalance techniques like SMOTE-Tomek Links, ADASYN, or cost-sensitive learning. On the other hand, using GEM self-reported survey data may not only introduce subjectivity but also cultural bias and measurement error which would not be detectable by the models. One way to achieve more profound analysis is to rely on administrative data or longitudinal panel surveys. Year-wise volatility in India also implies the existence of some unobserved macroeconomic variables; such as inflation, credit availability, digital adoption, or labour-market shocks that were not considered in the models. Incorporation of external economic indicators could help make the model more reliable and comprehensible. Also, giving more attention to the model's performance could lead to more precise decomposition of the drivers of OME and NME through the use of additional feature engineering values. Lastly, the addition of hybrid deep-learning and boosting frameworks or causal ML to the existing models could provide more detailed insights, especially in the case of policy-oriented entrepreneurship research.

7. Conclusion

This research examined the predictive accuracy of several supervised machine learning models for opportunity and necessity driven entrepreneurship (OME and NME respectively) using the GEM data for India and the global sample during the years 2015-2020. The results across all the analyses consistently show that tree-based ensemble models especially Random Forest, XGBoost, CatBoost, and H2O GBM considerably outperform the linear and distance-based models in terms of accuracy, discrimination power, and stability.

This indicates that entrepreneurial behavior is inherently complex and the result of non-linear interactions among the perceptions, socio-economic factors, and contextual conditions. The pooled results illustrate model dominance patterns quite distinctly: XGBoost is the strongest classifier for global OME, whereas Random Forest has the best performance for both OME and NME in India specifically.

However, the year-wise analysis brings out major nuances. Indian datasets are characterized by significant volatility from one year to the next, and the leadership changes between ensemble models, linear regressions and even probabilistic classifiers depending on the economic cycles, labour-market conditions, and contextual uncertainty.

On the contrary, global datasets are much more stable, with XGBoost and H2O GBM taking turns as top performers but consistently achieving high accuracy and AUC across all years. This shows that while pooled results reflect global entrepreneurial behavior accurately, they obscure meaningful temporal variations in India.

Appendix A

A1. Feature Interpretation

Independent Variables

Feature (abbreviation)	Description	Type
Respondent ID (SETID)	Unique identifier for each respondent	Identifier
Country Code (country)	Country code of respondent	Categorical
Survey Year (year)	Year in which the GEM survey was conducted	Categorical / Time
Household Income Category (GEMHHINC)	Household income category as reported in GEM dataset	Categorical
Educational Attainment (GEMEDUC)	Level of education attained (GEM-defined categories)	Categorical
Knowing an Entrepreneur (knowent)	Knows someone personally who started a business (1 = Yes; 0 = No)	Binary
Perceived Opportunities (opport)	Sees good opportunities to start a business (1 = Yes; 0 = No)	Binary
Perceived Skills (suskil)	Has skills/knowledge/experience to start a business (1 = Yes; 0 = No)	Binary
Fear of Failure (fearfail)	Fear of failure would prevent business startup (1 = Yes; 0 = No)	Binary
Media Attention to Entrepreneurship (nbmedia)	Media gives attention to entrepreneurship (1 = Yes; 0 = No)	Binary
Future Support for Entrepreneurship (futsup)	Expects to start a business in next 3 years (1 = Yes; 0 = No)	Binary
Full-Time Employment (occufull)	Employed full-time (1 = Yes; 0 = No)	Binary
Part-Time Employment (occupart)	Employed part-time (1 = Yes; 0 = No)	Binary
Job-Seeking Status (occuseek)	Currently looking for work (1 = Yes; 0 = No)	Binary
Student Status (occustu)	Current student (1 = Yes; 0 = No)	Binary

Home Duties (occuhome)	Engaged in home duties (1 = Yes; 0 = No)	Binary
Business Ownership (occurd)	Owner of a business (1 = Yes; 0 = No)	Binary
Gender (gender)	Respondent gender (1 = Male; 2 = Female)	Binary
Age (age)	Respondent age (in years)	Continuous
Household Size (hhsiz)	Number of individuals living in household	Continuous

Dependent Variables

FEATURE (abbreviation)	Description	Type
Opportunity-Driven TEA (TEAyyOPP)	Opportunity-driven Total Early-stage Entrepreneurial Activity (1 = Yes; 0 = No)	Binary
TEA – Necessity Entrepreneurship (TEAyyNEC)	Necessity-driven TEA (1 = Yes; 0 = No)	Binary

A2. NME India Year-Wise Model with Best AUC Results(2015-2020)

Year	Best AUC Model	Best AUC	Accuracy of Same Model
2015	CatBoost	0.826568	0.981884
2016	CatBoost	0.857592	0.971487
2017	Naive Bayes	0.863040	0.845692
2018	Naive Bayes	0.738069	0.761905
2019	Ridge	0.773578	0.832421
2020	CatBoost	0.806697	0.949749

A3. NME World Year Wise Model with Best AUC Results(2015-20)

Year	Best AUC Model	AUC	Accuracy
2015	XGBoost	0.793356	0.949112
2016	H2O GBM	0.795726	0.959684

2017	H2O GBM	0.819944	0.959153
2018	H2O GBM	0.793530	0.952343
2019	XGBoost	0.827326	0.867065
2020	H2O GBM	0.835518	0.863427

A4. OME India Year Wise Model with Best AUC Results (2015-2020)

Year	Best AUC Model	AUC	Accuracy
2015	Random Forest	0.881142	0.920290
2016	Ridge	0.819093	0.855102
2017	Random Forest	0.966005	0.981366
2018	Random Forest	0.860840	0.944099
2019	Ridge	0.778783	0.862667
2020	Lasso	0.795455	0.801508

A5. OME World Year Wise Model with Best AUC Results (2015-2020)

Year	Best AUC Model	AUC	Accuracy
2015	XGBoost	0.849005	0.887055
2016	H2O GBM	0.855802	0.895492
2017	XGBoost	0.851432	0.898868
2018	H2O GBM	0.842545	0.889121
2019	XGBoost	0.832664	0.885871
2020	H2O GBM	0.846255	0.882088

A6. NME INDIA 2015-20 POOLED CONFUSION MATRIX

Model	Class	TP	FP	TN	FN
Lasso	Yes	123	417	2275	60
Lasso	No	2275	60	123	417
Ridge	Yes	120	413	2279	63
Ridge	No	2279	63	120	413
Elastic Net	Yes	122	412	2280	61
Elastic Net	No	2280	61	122	412
Random Forest	Yes	54	35	2657	129
Random Forest	No	2657	129	54	35
ANN	Yes	0	0	2692	183
ANN	No	2692	183	0	0
KNN	Yes	76	745	1967	107
KNN	No	1947	107	76	745
XGBoost	Yes	53	55	2637	130

XGBoost	No	2637	130	53	55
Decision Tree	Yes	80	220	2472	103
Decision Tree	No	2472	103	80	220
Logistic Regression	Yes	69	312	2380	114
Logistic Regression	No	2380	114	69	312
Naive Bayes	Yes	111	553	2139	72
Naive Bayes	No	2139	72	111	553
CatBoost	Yes	56	45	2647	127
CatBoost	No	2647	127	56	45
H2O GBM	Yes	77	94	2598	106

A7. NME WORLD 2015-20 POOLED CONFUSION MATRIX

Model	Class	TP	FP	TN	FN
Lasso	Yes	170	1257	10675	362
Lasso	No	10675	362	170	1257
Ridge	Yes	179	1332	10600	353
Ridge	No	10600	353	179	1332
Elastic Net	Yes	176	1290	10642	356
Elastic Net	No	10642	356	176	1290
Random Forest	Yes	4	0	11928	528
Random Forest	No	11928	528	4	0
ANN	Yes	0	0	11932	532
ANN	No	11932	532	0	0
KNN	Yes	134	1368	10564	398
KNN	No	10564	398	134	1368
XGBoost	Yes	11	18	11914	521
XGBoost	No	11914	521	11	18
Decision Tree	Yes	1	0	11932	532
Decision Tree	No	11932	532	1	0
Logistic Regression	Yes	192	1448	10484	340

Logistic Regression	No	10484	340	192	1448
Naive Bayes	Yes	361	3909	8023	171
Naive Bayes	No	8023	171	361	3909
CatBoost	Yes	0	0	11932	532
CatBoost	No	11932	532	0	0
H2O GBM	Yes	34	96	11836	498
H2O GBM	No	11836	498	34	96

**A8. OME INDIA 2015-20
CONFUSION MATRIX**

Model	Class	TP	FP	TN	FN
Lasso	Yes	175	502	1644	58
Lasso	No	1644	58	175	502
Ridge	Yes	170	486	1660	63
Ridge	No	1660	63	170	486
Elastic Net	Yes	175	490	1656	58
Elastic Net	No	1656	58	175	490
Random Forest	Yes	54	48	2098	179
Random Forest	No	2098	179	54	48
ANN	Yes	0	0	2146	233

ANN	No	2146	233	0	0
kNN	Yes	87	683	1463	146
kNN	No	1463	146	87	683
XGBoost	Yes	53	52	2094	180
XGBoost	No	2094	180	53	52
Decision Tree	Yes	110	248	1898	123
Decision Tree	No	1898	123	110	248
Logistic Regression	Yes	172	483	1663	61
Logistic Regression	No	1663	61	172	483
Naive Bayes	Yes	150	480	1666	83
Naive Bayes	No	1666	83	150	480
CatBoost	Yes	65	71	2075	168
CatBoost	No	2075	168	65	71
H2O GBM	Yes	72	89	2057	161
H2O GBM	No	2057	161	72	89

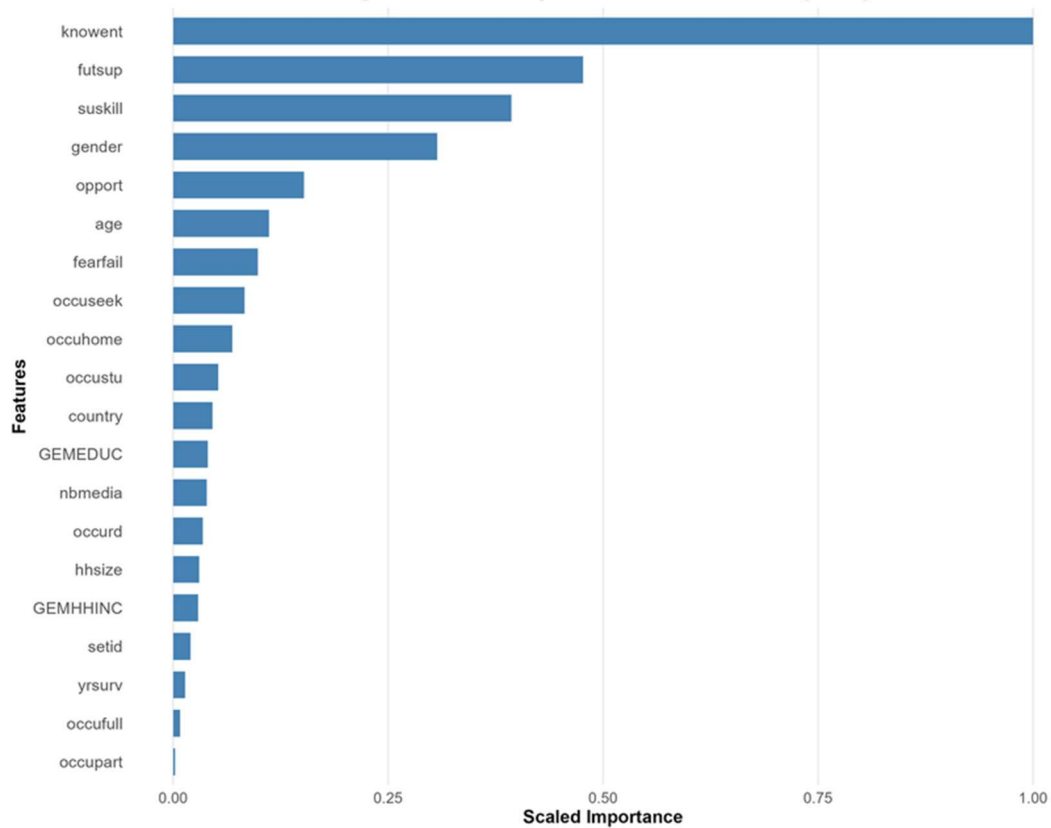
A9. OME WORLD 2015-20 CONFUSION MATRIX

Model	Class	TP	FP	TN	FN
Lasso	Yes	5845	19012	44746	1902
Lasso	No	44746	1902	5845	19012

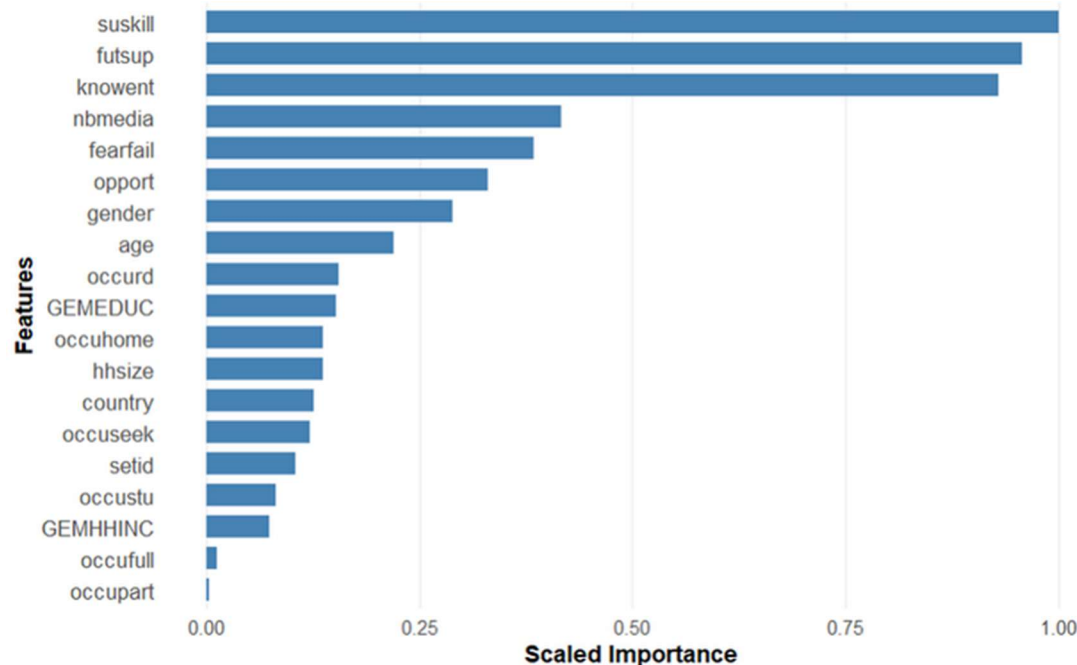
Ridge	Yes	5836	18634	45124	1911
Ridge	No	45124	1911	5836	18634
Elastic Net	Yes	5846	18740	45018	1901
Elastic Net	No	45018	1901	5846	18740
Random Forest	Yes	1078	1042	62716	6669
Random Forest	No	62716	6669	1078	1042
ANN	Yes	7747	63758	0	0
ANN	No	0	0	7747	63758
kNN	Yes	3986	16159	47599	3761
kNN	No	47599	3761	3986	16159
XGBoost	Yes	1111	952	62806	6636
XGBoost	No	62806	6636	1111	952
Decision Tree	Yes	4001	11394	52364	3746
Decision Tree	No	52364	3746	4001	11394
Logistic Regression	Yes	5841	18375	45383	1906
Logistic Regression	No	45383	1906	5841	18375
Naive Bayes	Yes	7606	62073	1685	141

Naive Bayes	No	1685	141	7606	62073
CatBoost	Yes	1289	1762	61996	6458
CatBoost	No	61996	6458	1289	1762
H2O GBM	Yes	1334	1580	62178	6413
H2O GBM	No	62178	6413	1334	1580

A10. Random Forest Feature Prediction OME World(2015-2020)



A11. Random Forest Feature Prediction NME World(2015-2020)



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